4.4.4 Production of induced emf by changing relative orientation of the coil with the magnetic field

Consider a rectangular coil of N turns kept in a uniform magnetic field $\vec{B}$ as shown in Figure 4.24. The coil rotates in anti-clockwise direction with an angular velocity ω about an axis, perpendicular to the field and to the plane of the paper.

At time $t = 0$, the plane of the coil is perpendicular to the field and the flux linked with the coil has its maximum value $\Phi_m = NBA$ (where A is the area of the coil).

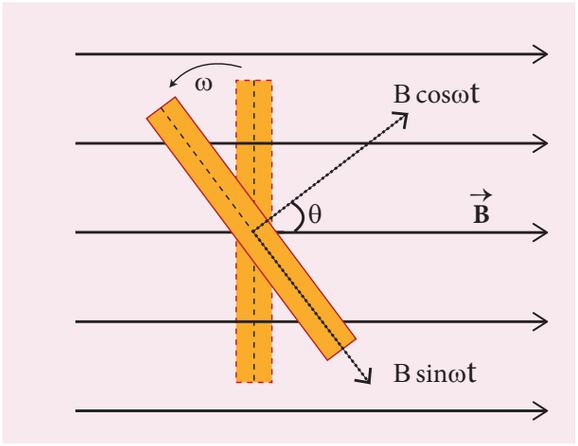


Figure 4.24 The coil has rotated through an angle $\theta = \omega t$

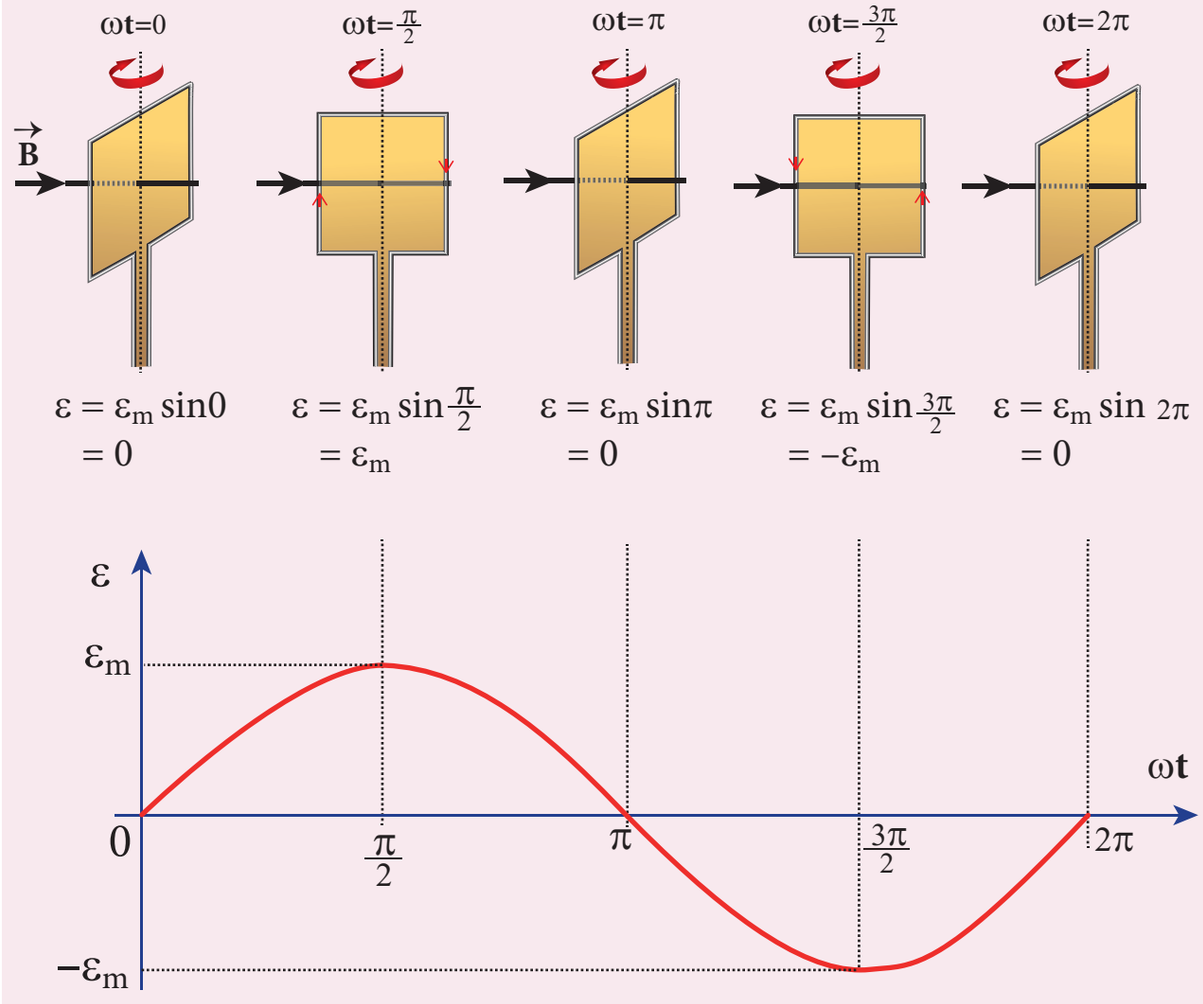


Figure 4.25 Variation of induced emf as a function of ωt

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